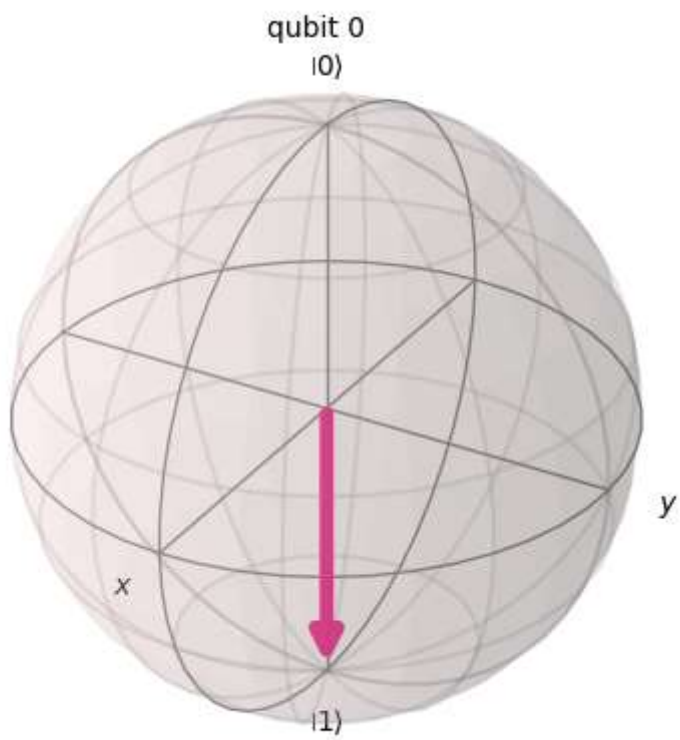
In [2]:
```
# Importing standard Qiskit libraries
from qiskit import QuantumCircuit,transpile
from qiskit.visualization import *
# qiskit-ibmq-provider has been deprecated.
# Please see the Migration Guides in https://ibm.biz/provider_migration_guide for m
from qiskit_ibm_runtime import QiskitRuntimeService, Sampler, Estimator, Session, O
# Loading your IBM Quantum account(s)
service = QiskitRuntimeService(channel="ibm_cloud")
from qiskit_aer import AerSimulator
from qiskit_ibm_runtime import QiskitRuntimeService
# Invoke a primitive. For more details see https://docs.quantum.ibm.com/run/primiti
# result = Sampler().run(circuits).result()
import numpy as np
# Importing standard Qiskit libraries
from qiskit_aer import Aer
#from qiskit import QuantumCircuit, transpile, assemble, IBMQ, execute
from qiskit.quantum_info import Statevector
from qiskit.visualization import plot_bloch_multivector, plot_histogram
```

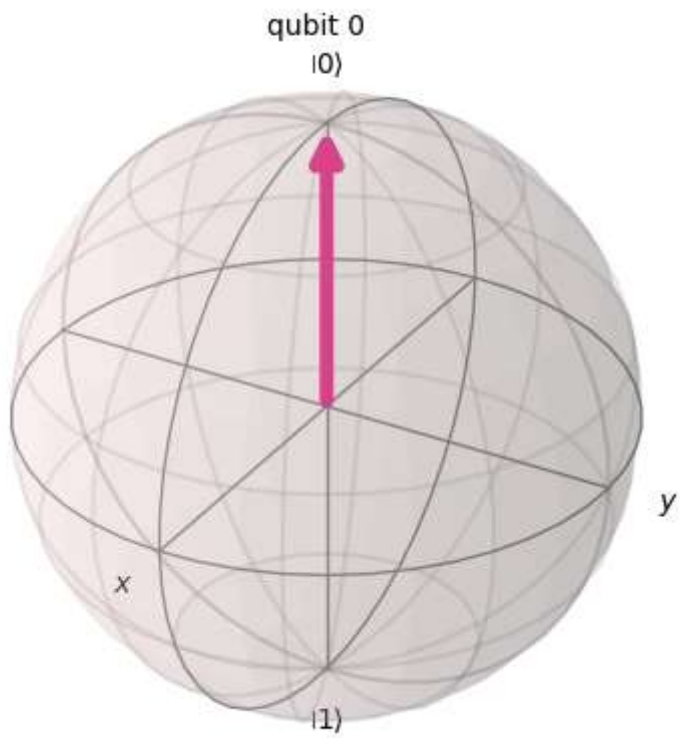
In [3]:
```
def lab1_ex1():
    qc = QuantumCircuit(1)

    qc.x(0)

    return qc

state = Statevector.from_instruction(lab1_ex1())
plot_bloch_multivector(state)
```

Out[3]:



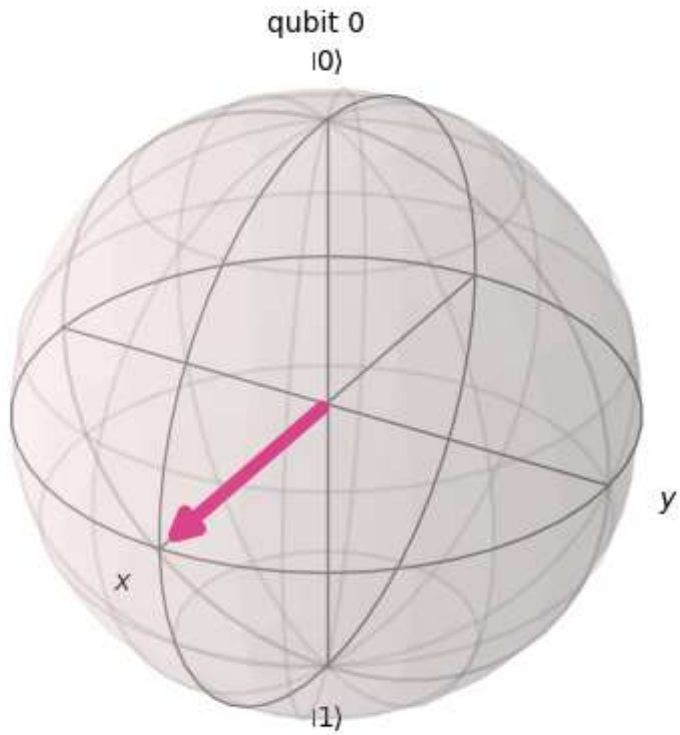
In [4]:
```
def lab1_ex1():
    qc = QuantumCircuit(1)

    qc.x(0)
    qc.x(0)

    return qc

state = Statevector.from_instruction(lab1_ex1())
plot_bloch_multivector(state)
```

Out[4]:



In [5]:
```
def lab1_ex2():
    qc = QuantumCircuit(1)

    qc.h(0)

    return qc

state = Statevector.from_instruction(lab1_ex2())
plot_bloch_multivector(state)
```

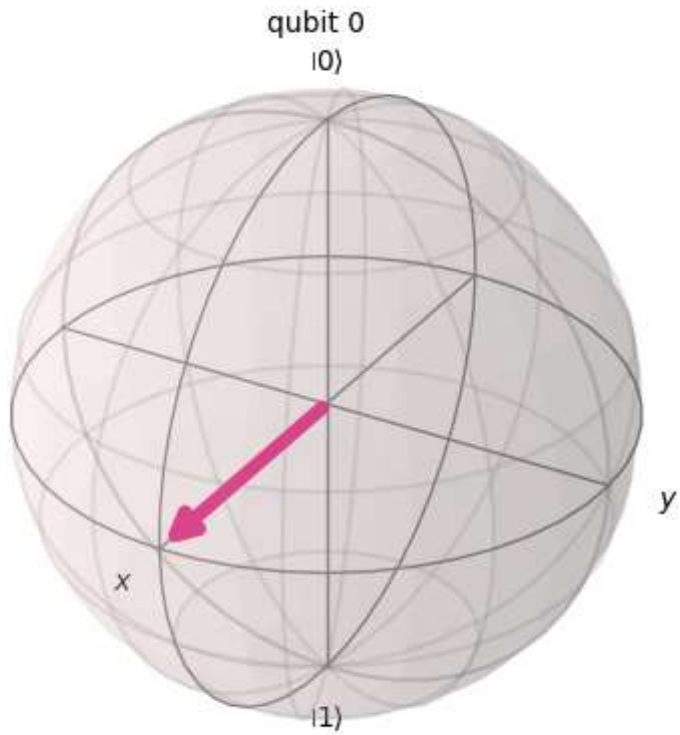
Out[5]:



In [6]:

```
def lab1_ex3():
    qc = QuantumCircuit(1)
    #
    #



    qc.h(0)

    #
    #
    return qc

state = Statevector.from_instruction(lab1_ex3())
plot_bloch_multivector(state)
```

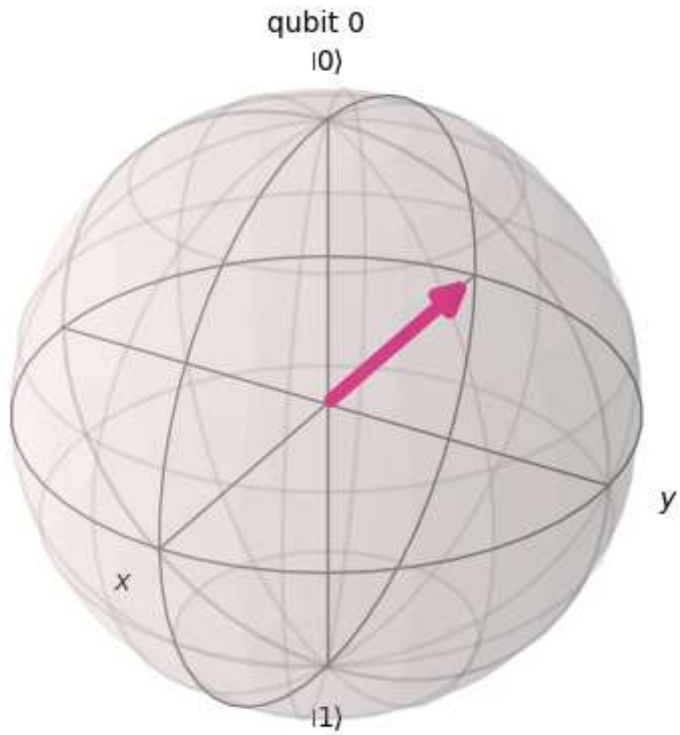
Out[6]:



In [7]:

```
def lab1_ex4():
    qc = QuantumCircuit(1)
    #
    #
    qc.x(0)
    qc.h(0)
    #
    #
    return qc

state = Statevector.from_instruction(lab1_ex4())
plot_bloch_multivector(state)
```

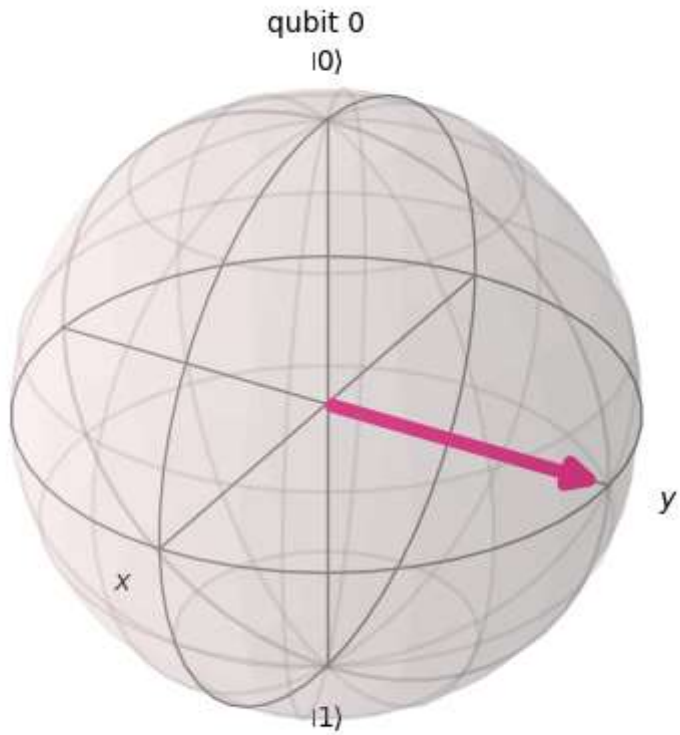
Out[7]:



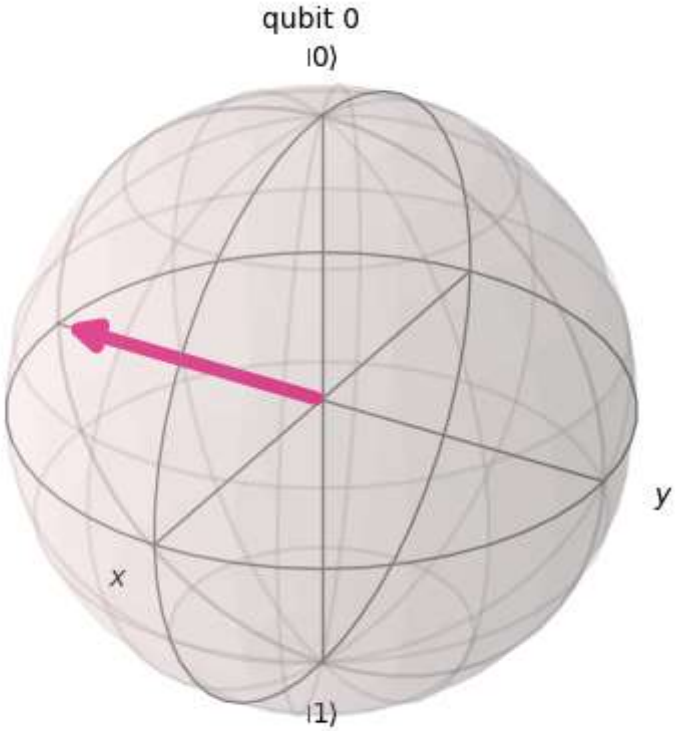
In [8]:

```
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector
from qiskit.visualization import plot_bloch_multivector
import matplotlib.pyplot as plt

# Create 1-qubit circuit
qc = QuantumCircuit(1)

# Prepare |+i⟩
#Student please write your code here for |+i⟩
qc.h(0)
qc.s(0)
state = Statevector.from_instruction(qc)
plot_bloch_multivector(state)
```

Out[8]:



In [9]:

```
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector
from qiskit.visualization import plot_bloch_multivector
import matplotlib.pyplot as plt

# Create 1-qubit circuit
qc = QuantumCircuit(1)

# Prepare |-i⟩
#Student please write your code here for |-i⟩
qc.h(0)
qc.sdg(0)
state = Statevector.from_instruction(qc)
plot_bloch_multivector(state)
```

Out[9]: